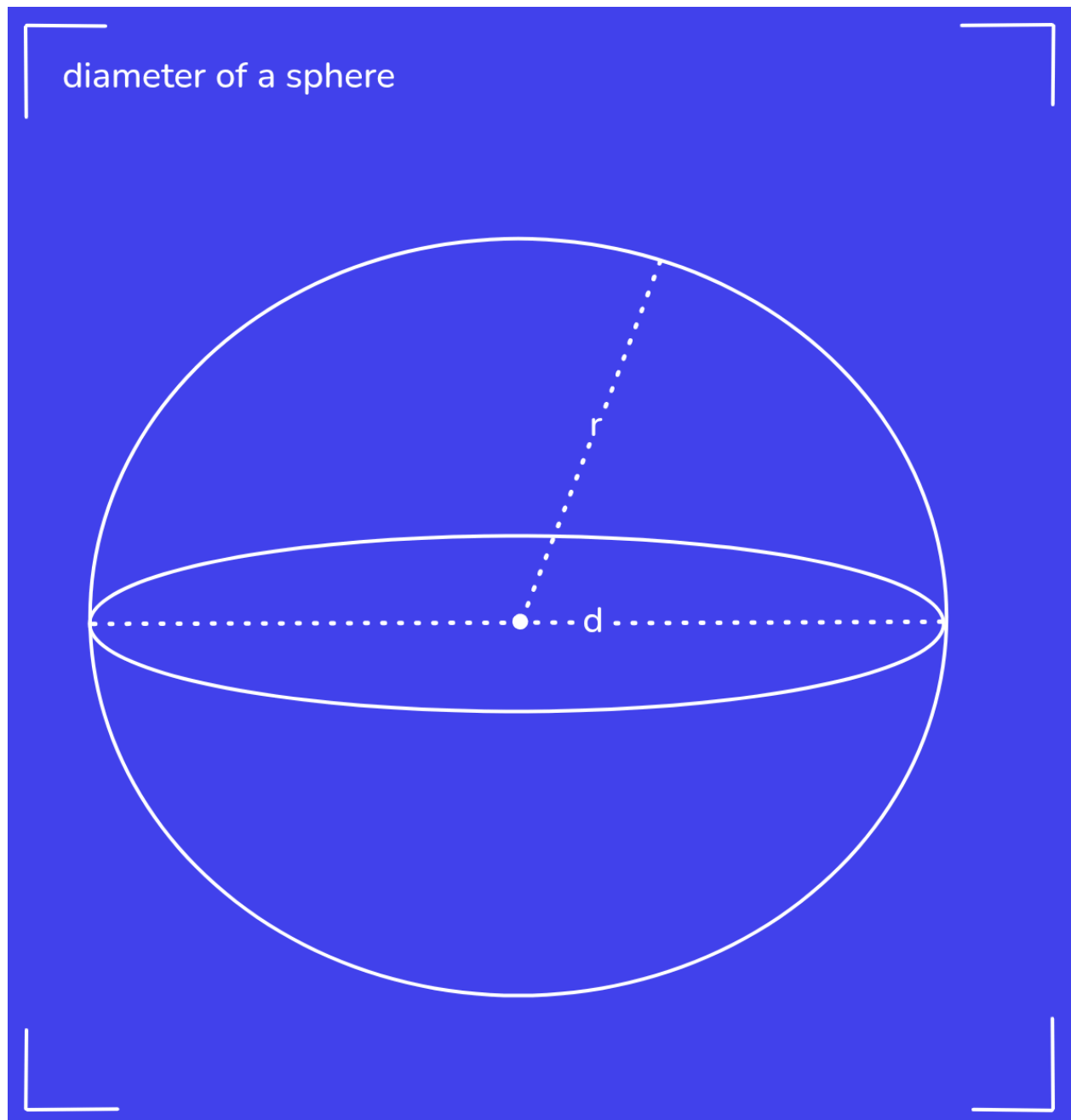




Ray-Sphere Intersection



A Sphere is defined by:

- A center point "C"
- A radius "R"

A point "P" is on the sphere if:

$$|P - C| = R$$

The Intersection Test, whose steps are described below, consists in determining if the path of a ray crosses with the surface of the sphere.

Step 1: What do we Know



'.' is the symbol for the Dot Product. Also known as Escalar Product or Internal Product.

It means that the product multiplies each individual property of both factors to get an escalar result.

For example, for two 3D vectors "D" and "V" whose X, Y, and Z properties are:

D = (1, 0, 0) (pointing right along x-axis)

V = (3, 4, 0) (pointing somewhere)

$V \cdot D = (3 \times 1) + (4 \times 0) + (0 \times 0) = 3$

The Dot Product returns a value of "3", which is an Escalar Magnitude.

- Sphere
 - Center point "C"
 - Radius "R"

- Any point "P" on sphere surface satisfies $|P - C| = R$
- Ray
 - Origin "O" (position of the camera)
 - Direction "D" (unit vector)
 - Any point along ray $P = O + t \cdot D$ where " $t > 0$ "

Step 2: What are we Trying to Find

We want to find values of " t " where the ray point equals a sphere surface point:

$$\begin{aligned} \text{Ray point} &= \text{Sphere surface point} \\ O + t \cdot D &= P \quad \text{where} \quad |P - C| = R \end{aligned}$$

Step 3: Substitute

Replace "P" in the sphere equation with " $O + t \cdot D$ ":

$$|(O + t \cdot D) - C| = R$$

This reads: "The distance from (the ray point) to (the sphere center) equals the radius".

Step 4: Square Both Sides



A Quadratic Equation is just a fancy name for an equation that involves a Squared Term (something x itself).

In 3D space, distance is calculated using the Pythagorean Theorem (the same one that's used to find the hypotenuse of a triangle):

$$Distance = \sqrt{(x^2 + y^2 + z^2)}$$

Where (x, y, z) is the vector from C to P.

So, in the following equation we would have a square root because of the Pythagorean Theorem, but we square both sides to remove it.

This leaves "R" squared, and the rest of the formula is free of square roots, which is simpler to manage.

$$|(O - C) + t \cdot D|^2 = R^2$$

Let "V = O - C" (vector from sphere center to ray origin):

$$|V + t \cdot D|^2 = R^2$$



We get a quadratic because we're finding where a straight line (the ray) meets a curved surface (the sphere).

The square appears because we're using the Pythagorean Theorem (distance formula).

Step 5: Expand Dot Products

For any vector "A": $|A|^2 = A \cdot A$

$$(V + t \cdot D) \cdot (V + t \cdot D) = R^2$$

$$V \cdot V + 2t(V \cdot D) + t^2(D \cdot D) = R^2$$

Step 6: Rearrange Into Quadratic Form



Note that we set one side of the equation to zero.

Considering that an equation is equal because both sides weight the same:

- When we have "something = something_else"
- We can rearrange it to "something - something_else = 0"

This is valid because if two things are equal, subtracting them returns zero.

In the following example, "R" is moved to the left side of the equation as a negative value.

$$t^2(D \cdot D) + 2t(V \cdot D) + (V \cdot V - R^2) = 0$$

This becomes a quadratic equation: $at^2 + bt + c = 0$

Where:

- $a = D \cdot D$ (dot product of direction)

- $b = 2 \times (V \cdot D)$
- $c = V \cdot V - R^2$



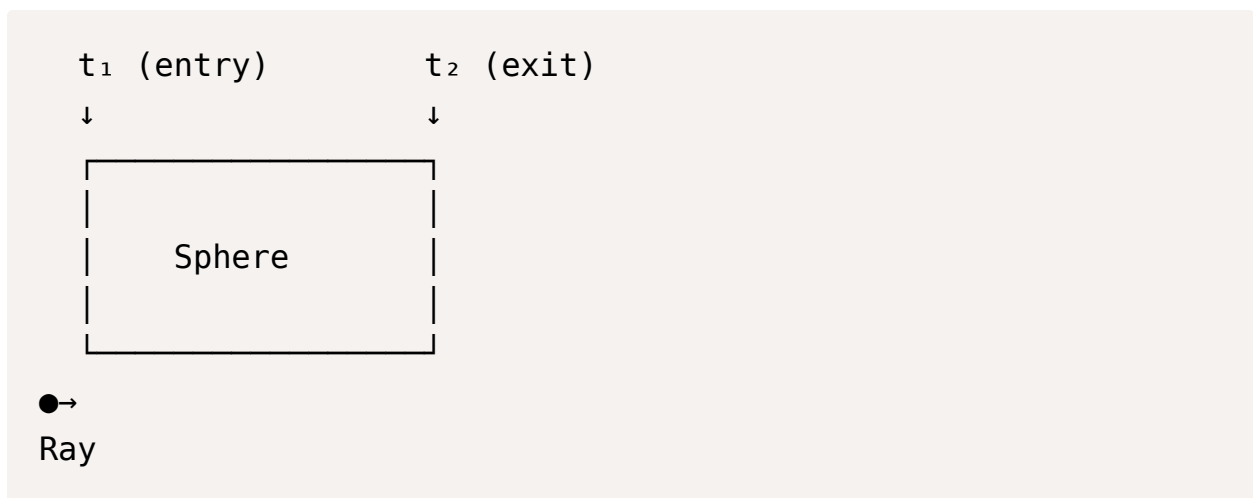
Quadratic Equations have a standard form that mathematicians have already solved:

- $at^2 + bt + c = 0$

This is like having a universal key that fits any lock of this shape. By rearranging our equation to match this pattern, we can use the Quadratic Formula to unlock the answer.

What This Equation Obtains

This quadratic gives us up to two solutions for "t":



- " t_1 " is where the ray enters the sphere (smaller positive value)
- " t_2 " is where the ray exits the sphere (larger positive value)

Discriminant



The Discriminant is the part that determines one of many possible solutions.

It "discriminates" (distinguishes) between these three cases.

The name literally means "the one that tells them apart".

The Discriminant $\Delta = b^2 - 4ac$ tells us:

- If " $\Delta < 0$ ": No intersection, the ray misses
- If " $\Delta = 0$ ": Ray touches sphere at one point (tangent)
- If " $\Delta > 0$ ": Ray hits sphere at two points (entry and exit)

The smallest positive "t" is the Intersection Point.